Avni Kothari

Senior Applied Scientist

- I am a Senior Applied Scientist with 6 years of software and ML engineering experience in AI healthcare and AI safety
- My ML engineering experience focuses on: 1) building, evaluating, and improving production ML models, 2) translating ambiguous projects into products and measurable metrics to inform business decisions, and 3) designing rigorous offline experiments and LLM evaluations to inform production systems
- My AI research, presented at **ICLR, AAAI, NeurIPS, & NPJ**, focuses on: 1) improving transparency and alignment in high-stakes AI models, 2) ensuring fairness by detecting scenarios of preclusion & 3) ensuring that AI benefits all individuals by creating LLM tools for those in resource-constrained settings

WORK EXPERIENCE

Senior Applied Scientist; Salesforce; SF, CA — Jan 2026 – Present

- Led the evaluation and mitigation of Agentforce's truthfulness and neutrality across 1,000+ custom-generated prompt-response pairs, translating vague government procurement language into concrete compliance metrics and using human annotation to validate LLM-as-a-judge results
- Designed experimental conditions to test mitigation strategies, identifying an approach that improved truthfulness and neutrality while achieving a 15% statistically significant reduction in miss rates, presenting directly to executive leadership
- Designed a reproducible testing repository and conducted a comprehensive codebase audit for the Agentforce reasoning engine to evaluate safety system prompt necessity, leading 8+ collaborative sessions with the partner team and cross-functional stakeholders
- Collaborated with a vendor to design 2,000+ multi-turn adversarial attacks across 10+ ethical risk dimensions for single-agent injections and multi-agent miscoordination, conflict, and collusion attack strategies
- Directed a research intern in delivering a comprehensive evaluation report on 4 asynchronous safety guardrails, assessing their viability for complementary approaches and measuring single- and multi-agent miss and false alarm rates

Research Engineer; University of California, SF; SF, CA — Sept 2023 – Dec 2025

- Led end-to-end development and architecture of a GPT-4o-powered patient summary tool that processes 200+ clinical notes to extract patient social needs and discharge information for 15+ clinical social workers
- Conducted evaluation of GPT-4o-generated patient summaries from 200+ clinical notes using LLM-as-a-judge methods while overseeing and training human annotators
- Developed a multimodal method to improve interpretability of black box models by generating and refining concept bottleneck models within a Bayesian framework for uncertainty quantification, outperforming baseline methods by 35%
- Created a method to use a Bayesian tree based model that merges Llama 2's domain knowledge with empirical data, increasing interpretability by 40%, as confirmed by domain experts, while matching top predictive performance and quantifying risk
- Designed and deployed a scalable ETL pipeline to process health record data, enabling ML training and evaluation for 3K+ patient records and 30K+ patient visits
- Built, deployed, and evaluated a custom clinical risk prediction model adopted by 10+ clinics, achieving 12% higher accuracy than the general clinical model

Software Engineer; Edovo; Chicago, IL — June 2020 – May 2021

- Architected, tested, and deployed an educational content platform using Elasticsearch to handle 700K+ requests per day
- Led 10+ requirement gathering sessions with Product owners to re-build a platform
- Created a data pipeline and job to merge 4B rows of user event data in PostgreSQL

Lead Software Engineer; 8th Light; Chicago, IL — Jan 2017 – Mar 2019

- Implemented and deployed a scalable load testing platform simulating 1000+ RPS
- Engineered API integrations to sync 1000+ interactions/ minute in different timezones

PAPERS

MAStrike: Shapley-Guided Collusive Red-Teaming on Multi-Agent Systems

CONTACT

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LinkedIn | Github
Website | Google Scholar

SKILLS

ML Engineering Skills:

- ML Pipelines and ETL
- Multimodal datasets
- ML Deployment
- ML Development & Evaluation
- EHR/EMR Data Processing

AI Research Skills:

- Multi-Agent Safety
- Adversarial Red-Teaming
- Interpretability
- Robustness
- Fairness and Bias
- Risk Quantification
- LLMs and Foundation Models
- LLM Evaluations
- Human-in-the-Loop
- Clinical NLP

Tools and Frameworks:

- Python (Hugging Face, Pytorch, Scikit-learn, Numpy, Pandas)
- AWS (EC2, S3, Terraform, Deployment Strategies)
- Elasticsearch
- DB: SQL, Postgres, DuckDB
- Docker

EDUCATION

University of California, San Diego
Masters in Computer Science
Thesis: Foundations of Model Agnostic Recourse Verification
San Diego, CA — 2021 - 2023

University of Texas at Austin
Bachelors in Mathematics and Economics

Chejian Xu, Zhaorun Chen, Jingyang Zhang, Freddy Lecue, Avni Kothari, et al; *Pre-print, 2026*

Human-AI Co-design for Clinical Prediction Models
Jean Feng*, Avni Kothari*, et al; *NPJ Digital Medicine, 2026*
** denotes equal contribution*

When the Domain Expert Has No Time and the LLM Developer Has No Clinical Expertise: Real-World Lessons from LLM Co-Design in a Safety-Net Hospital
Avni Kothari, et al; ***AAAI - Oral Presentation**, 2026*

Bayesian Concept Bottleneck Models with LLM Priors
Jean Feng, Avni Kothari, et al; *NeurIPS, 2025*

Prediction Without Preclusion: Recourse Verification With Reachable Sets
Avni Kothari, et al; ***ICLR – Top 5% among submissions**, 2024*

Implementing a Predictive Model to Reduce Hospital Readmissions in a Safety Net Healthcare System
Arturo Gasga, Avni Kothari, et al; *ML4H - Machine Learning for Health, 2024* ***Oral Spotlight***

RESEARCH SOFTWARE

bc-llm — June 2024

- Implemented and developed a multimodal method using Metropolis Gibbs sampling to identify interpretable features for complex models
- Benchmarked and implemented 5+ comparator methods against our method, achieving performance comparable to or exceeding black-box models

reachml — June 2022

- Constructed a Mixed Integer Program to handle 50+ feature constraints for counterfactual explanations and test for robustness
- Created a model-agnostic fairness and safety audit to identify scenarios of preclusion
- Developed an HPC-based experimental pipeline to audit 200K+ individuals and benchmark results against baseline methods

Minor: Computer Science
Austin, Texas — 2011 - 2016

POSTER PRESENTATIONS

- **NeurIPS** Workshop Statistical Foundations of LLMs and Foundation Models — Dec 2024
- **ICML** Workshop on Spurious Correlations, Invariance and Stability — Jul 2023
- **ICML** Workshop on Data-centric ML Research — Jul 2023

TEACHING EXPERIENCE

- Teaching Assistant for Interpretability & Explainability in ML — UC San Diego — Fall 2022